ALVIN LEE

alvinlee225@gmail.com | (908) 938-8937 | Portfolio | GitHub | LinkedIn

EXPERIENCE

Blueflame AI Jan 2026 – Present

Software Engineer

- Led Blueflame's side of the account-merge design with Grata for a joint launch, defining how Grata-only, Blueflame-only and shared users are matched across two separate backends and keep their existing logins, through two weeks of cross-company design reviews.
- Designed write-once identity linking keyed on Grata's stable user ID that refuses conflicts instead of overwriting, so a bad match can never attach one person's data to another account.
- Defined field ownership and write ordering between the systems: one owner per shared field, revocations sync but grants never do, seat limits commit locally only after Grata accepts, and US/EU data stays in its region.
- Led Blueflame's pushback in cross-team design disputes, narrowing a proposed cross-system user service to three operations and holding Grata's org-mapping changes until safety fixes were adopted.
- Served as Blueflame's primary contact for launch identity incidents, tracing duplicate accounts to a cross-company retry/timeout race by correlating AWS and Datadog logs.
- Designed and shipped Datasite VDR sharing with a fail-closed, least-privilege access model enforced across search, RAG and code execution, rolled out through a staged backfill of 142 accounts and ~15K workspaces.
- Architected the permission foundation for Blueprints V3 (groups, ACLs, atomic bulk operations), validated at 10K blueprints and against production-scale migrations.

PepsiCo Jul 2023 – Jan 2026

Software Engineer, Data Platform

- Built self-service Snowflake provisioning (Jira, GitHub, Airflow), cutting turnaround from 2 days to 1 hour.
- Migrated 200+ service accounts to RSA key auth with zero downtime, backed by automated key rotation in Terraform and Azure Key Vault.
- Served as global Terraform SME, building modules for 5 multi-region Snowflake accounts that cut provisioning time 60%; mentored 15+ engineers.

Software Engineering Intern, Data Platform May 2022 – Aug 2022

- Built pipeline monitoring and alerting in Python and SQL that reduced MTTR by ~40%.

PROJECTS

Snowflake Analytics MCP Server *Python, FastMCP, pandas*

- Built an MCP server with 15+ tools for Snowflake cost, security and anomaly analysis.

Syncify *FastAPI, React, OAuth 2.0*

- Built a Spotify-to-Apple Music migration service with ~95% track match accuracy.

TECHNICAL SKILLS

Languages: Python, SQL
Systems: Distributed systems, system design, access control (RBAC, ACLs), REST APIs, RAG, LLM tooling
Infrastructure: AWS, Azure, Terraform, Docker, Kubernetes
Data: MongoDB, DocumentDB, Redis, Snowflake, dbt, Airflow
Identity: Auth0, OAuth 2.0, OIDC, JWT

EDUCATION

New York University May 2023

B.A. in Mathematics and Computer Science